Ordinals and Typed Lambda Calculus
Defining Ordinals in Martin-Löf Type Theory

Andrés Sicard-Ramírez

Universidad EAFIT

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Hilbert Ordinals

Definition

Hilbert ([1925] 1967) (originally published as (Hilbert 1926)) axiomatically defined the ordinals of the (cumulative) second number class. Let N and O be unary predicates representing natural and ordinal numbers, respectively.

$$\mathrm{N}\,\mathrm{zero},$$
$$\forall n\,[\mathrm{N}n \to \mathrm{N}(\mathrm{succ}(n))],$$
$$\{P\,\mathrm{zero} \wedge \forall n\,[Pn \to P(\mathrm{succ}(n)]\} \to \forall n\,(\mathrm{N}n \to Pn),$$

$$\mathrm{O}\,\mathrm{zero},$$
$$\forall n\,[\mathrm{O}n \to \mathrm{O}(\mathrm{succ}(n))],$$
$$\forall n\,[\mathrm{N}n \to \mathrm{O}(f(n))] \to \forall n\,[\mathrm{O}(\lim f(n))],$$
$$\{P\,\mathrm{zero} \wedge \forall n\,[Pn \to P(\mathrm{succ}(n))] \wedge \forall n\,[P(f(n)) \to P(\lim f(n)]\} \to \forall n\,(\mathrm{O}n \to Pn).$$

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Discussion

Can we define a data type for representing ordinal numbers from Hilbert's axioms?

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Remark

Cumulative second number class have been studied by various authors (see, e.g., (Church and Kleene 1937; Coquand, Hancock and Setzer 1997; Howard 1972; Martin-Löf 1984, [1970] 2001; Stenlund 1972)).

Martin-Löf's Ordinals

Remark

Martin-Löf (1984) defined ordinals in his type theory. He did not mention Hilbert but Cantor when given his definition.

Introduction rules

The introduction rules for Martin-Löf's ordinals are the following ones.

$$\frac{}{\mathsf{zero} : \mathsf{Nat}} \qquad \frac{n : \mathsf{Nat}}{\mathsf{succ}\, n : \mathsf{Nat}}$$

$$\frac{}{\mathsf{zero}_o : \mathsf{On}} \qquad \frac{n : \mathsf{On}}{\mathsf{succ}_o\, n : \mathsf{On}} \qquad \frac{f : \mathsf{Nat} \to \mathsf{On}}{\mathsf{lim}_o\, f : \mathsf{On}}$$

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